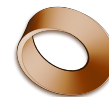


Name: _____

Class: _____

Mr. Rawson • GCSE Computer Science



Year 11 2026

Programming 5 — String Activities

Work down the page. You are not expected to reach the bottom.

Warm up

1 • Predict, then run adapted from Paul Long

Two strings for this one:

```
a = "Mississippi"
b = "Count every character!"
```

Write your prediction *before* you type anything.

ERL	Prediction	Actual
a.length		
b.length		
a.substring(0,4)		
b.left(5)		
b.right(10)		
b.substring(6,5)		

Count the spaces. They are characters too, and the exclamation mark counts as one as well.

Build

2 • Initials

Ask the user for their full name and print their initials.

Ada Augusta Lovelace should give AAL.

Start with two names and get it working, then worry about people with three.

3 • The race code adapted from Paul Long — the good one

A swimming gala gives every race a short code, built out of four pieces of information.

Variable	Value	What goes into the code
gender	"Boys"	the first 1 character
distance	"400"	all of it
event	"Relay"	the first 2 characters
age_group	"Under 13"	the first 1 , then the last 2

Work out the code on paper first. Then write the program that builds it.

The code is: _____

Then make it useful: ask the user for the four pieces rather than fixing them in the code, so it works for any race in the gala.

Check yourself on age_group. It is "Under 13" — **8** characters, and the space is one of them. You need the **first 1** and the **last 2**, and nothing in between.

4 • The part number adapted from Paul Long

A factory builds a part number from three things:

```
manuf_year = "2010"  
manuf_name = "Blobby Enterprises"  
stock_code = "0987654321"
```

The part number is **the last digit of the year**, then **the first 2 characters of the name**, then **the last 3 digits of the stock code**.

The part number is: _____

Note the capital B and the lower-case l. You are taking characters exactly as they are — no one asked you to change the case.

5 • Password checker

Ask the user for a password and tell them whether it is acceptable. It must be:

- at least 8 characters long
- **not** the word password, in any combination of upper and lower case

The second rule is the interesting one. PassWord and PASSWORD must both be rejected — and you are not going to list every combination.

Stretch

6 • The Caesar shift character codes doing real work

Julius Caesar encrypted his messages by moving every letter three places along the alphabet. A becomes D, B becomes E, and so on.

HELLO becomes KHOOR.

You already have everything you need: ASC turns a character into its number, CHR turns a number back into a character. Adding 3 in between is the whole cipher.

- Encrypt a single letter typed by the user.
- Encrypt a whole word. *You will need a loop.*
- Decrypt it again, and check you get the original back.

The problem you will hit. What happens to Z? Adding 3 takes you past the end of the alphabet into punctuation.

It needs to wrap round to the start. *Which operator from last lesson wraps things round?*

*One last thing, and then we will never mention it again. A **telephone number is a String**. Store 01382 as an Integer and the leading zero vanishes, because an Integer has no idea it was ever written down. GCSE Computer Science is extremely fond of this fact — expect to be asked about it more than once.*